A Solution for Algebraic Equations $x^2 + 1 = 0$ and $x^2 - 1 = 0$ is $\sqrt{-1}$

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Abstract: The imaginary number $\sqrt{-1}$ plays a vital in complex number system. This paper presents the imaginary number $\sqrt{-1}$ as solutions for algebraic equations which are discussed in it.

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1. Introduction

The square root of the positive one is ±1, but the negative one has no real square roots and $\sqrt{-1}$ is named as an imaginary number. When we try to find the solution for the algebraic equation $x^2 + 1 = 0$, the complex number came to the usage in the number systems. $\sqrt{-1}$ plays a vital in complex number system. Also, there are no complex numbers without $\sqrt{-1}$.

2. Algebraic Equations and its Solutions $\sqrt{-1}$

$i = \sqrt{-1}$ denote the imaginary number. Also, $i^2 = -1$. The equation $x^2 + 1 = 0$ is true if and only if $x^2 = i^2$, where $i^2 = -1$ and $i = \sqrt{-1}$. $a + ib$ is the general form of a complex number, where a and b are real numbers.

Lemma: $\sqrt{-1}$ is a solution for both equation $x^2 - 1 = 0$ and $x^2 + 1 = 0$.

Proof.

$1 = (-1) \times (-1)$ is absolutely true. $\sqrt{-1}\sqrt{-1} = \sqrt{-1}\sqrt{-1}$ is also true.

Then, $\dfrac{\sqrt{-1}\sqrt{-1}}{\sqrt{-1}\sqrt{-1}} = 1$, where $\sqrt{-1}\sqrt{-1} = \pm 1 \Rightarrow \left(\sqrt{-1}\right)^2 = \pm 1$.

$\therefore \left(\sqrt{-1}\right)^2 = +1$ or -1.

Now, $x^2 - 1 = 0 \Rightarrow \left(\sqrt{-1}\right)^2 - 1 = 0 \Rightarrow +1 - 1 = 0$.
$x^2 = \left(\sqrt{-1}\right)^2 \Rightarrow x = \sqrt{-1}$, where $x^2 = \left(\sqrt{-1}\right)^2 = +1$.

Next, $x^2 + 1 = 0 \Rightarrow \left(\sqrt{-1}\right)^2 + 1 = 0 \Rightarrow -1 + 1 = 0$.
$x^2 = \left(\sqrt{-1}\right)^2 \Rightarrow x = \sqrt{-1}$, where $x^2 = \left(\sqrt{-1}\right)^2 = -1$.

Hence, Lemma is proved.

3. Conclusion

The imaginary number $\sqrt{-1}$ plays a vital in complex number system. In this article, $\sqrt{-1}$ is showed as a solution for both algebraic equations $x^2 + 1 = 0$ and $x^2 - 1 = 0$.

References

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